

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

EXECUTIVE SUMMARY The analysis reveals that **haircare** products exhibit the highest average defect rate (2.48 %) while **cosmetics** have the lowest (1.92 %). **Carrier C** delivers the best revenue to shipping cost ratio (US \$1,138.56 per shipping dollar) despite a moderate defect rate (2.60 %). Thirty-six SKUs failed inspection, with several exceeding a 4 % defect rate, and three high-risk stockouts (e.g., **SKU34** with a risk score of 13) are imminent due to minimal inventory and long lead times.

KEY FINDINGS - **Product type defect hierarchy:** Haircare = 2.483 % > Skincare = 2.335 % > Cosmetics = 1.919 % (average defect rates). - **Top carrier efficiency:** Carrier C – \$184,880 revenue / \$162.38 cost = **\$1,138.56 per \$1 shipping** (highest ratio); followed by Carrier B (\$1,055.70) and Carrier A (\$917). - **Failed inspection volume:** 36 SKUs flagged; high-severity cases (>4 % defect) include **SKU42 (4.94 %)**, **SKU65 (4.91 %)**, **SKU50 (4.75 %)**, **SKU3 (4.75 %)**, and **SKU73 (4.62 %)**. - **Highest stockout risk:** **SKU34** (skincare) – stock = 1, lead time = 26 days, risk score = 13; **SKU68** (haircare) – stock = 0, lead time = 8 days, risk score = 8; **SKU24** (haircare) – stock = 4, lead time = 26 days, risk score = 5.2. - **Carrier vs. defect trade-off:** While Carrier C maximizes revenue per dollar, its average defect rate (2.60 %) exceeds Carrier B's (1.76 %). - **Transportation mode impact:** Air yields the lowest average defect rate (1.824 %) across 26 SKUs, whereas Road has the highest (2.621 %) across 29 SKUs.

RISKS AND RECOMMENDATIONS The combination of high defect rates in the haircare line and reliance on carriers with elevated defect rates (especially Carrier C) threatens product quality and brand reputation. Immediate actions: (1) prioritize corrective quality initiatives for the 36 failed inspection SKUs, focusing on those >4 % defect; (2) renegotiate carrier contracts to favor low defect, high efficiency partners (e.g., prioritize Carrier B where feasible); and (3) increase safety stock or expedite replenishment for the top three risk score SKUs (SKU34, SKU68, SKU24) to mitigate imminent stockouts.